

Reflections from the Field

Lessons Learned as a Robotics Trainer in Kenyan Schools

Prepared for portfolio publication

Author: Danton Makau

Focus: Robotics facilitation, learner engagement, equity, and practical STEM delivery

Year: 2026

Reflections from the Field

Lessons Learned as a Robotics Trainer in Kenyan Schools

Author: Danton Makau

Category: STEM Education / Robotics Training / Field Practice

Year: 2026

Abstract

Robotics education is often discussed through the language of innovation, competition, and technology. However, the most important lessons emerge in the classroom and workshop floor, where learners encounter robots for the first time, struggle with code, collaborate in teams, and slowly discover that technology is something they can create rather than merely consume. This white paper reflects on practical lessons learned from robotics facilitation in Kenyan schools and youth programs. It examines learner motivation, classroom management, access barriers, teacher support, gender inclusion, competition readiness, and the importance of connecting robotics to local problem-solving. The paper argues that successful robotics education depends less on the sophistication of the kit and more on the quality of facilitation, learner confidence, structured practice, and a culture that allows students to fail safely and improve continuously.

Executive Summary

Robotics programs can transform how learners understand science, technology, engineering, and mathematics. In practice, however, robotics training is not simply about building machines. It is about building confidence, teamwork, persistence, communication, and problem-solving habits. Field experience shows that learners are most engaged when activities are practical, locally relevant, and structured around clear challenges. Trainers must balance excitement with discipline, creativity with safety, and competition with learning. This paper provides field-based reflections and recommendations for schools, trainers, NGOs, and education partners seeking to implement meaningful robotics programs in Kenya and similar contexts.

Table of Contents

1. Introduction: Robotics Beyond the Kit
2. The First Encounter: Building Learner Confidence
3. What Learners Actually Gain from Robotics
4. Common Field Challenges
5. The Role of the Trainer
6. Girls in STEM and Inclusive Participation
7. From Workshops to Competitions
8. Recommendations for Sustainable Robotics Programs
9. Conclusion

References

1. Introduction: Robotics Beyond the Kit

In many conversations, robotics education is reduced to equipment: kits, sensors, motors, laptops, competitions, and software. These tools matter, but they are only part of the story. The real power of robotics appears when learners begin to understand that they can command a machine, test an idea, make a mistake, repair it, and try again. In that moment, a robot becomes more than a device; it becomes a learning partner.

Field experience in Kenyan schools shows that robotics can shift how learners view technology. Many students are familiar with phones, social media, and digital entertainment, but fewer have had the opportunity to create technology. Robotics changes this relationship. Learners move from users to builders. They begin to ask questions such as: Why did the robot turn too early? Why is the sensor reading wrong? How can we make it faster? What can we improve?

These questions are powerful because they place learners in an active role. Instead of waiting for answers, they investigate. Instead of copying notes, they experiment. Instead of fearing mistakes, they begin to see mistakes as part of learning. This is the deeper educational value of robotics.

However, robotics is not automatically transformative. A poorly planned session can become chaotic, confusing, or exclusive. Some learners may dominate the kit while others watch silently. Some may feel intimidated by programming. Some schools may treat robotics as a showcase activity rather than a learning process. For robotics to succeed, facilitation must be intentional.

2. The First Encounter: Building Learner Confidence

The first robotics session is often emotional. Learners may be excited, nervous, curious, or unsure of themselves. Some approach the kit confidently, while others hesitate because they believe technology is for a special group of people. The trainer's first task is therefore not technical; it is psychological. Learners must believe that they are allowed to touch, try, fail, and learn.

Confidence grows when the first activity is simple and achievable. A beginner session should not start with complex programming or competition strategy. It should begin with a small success: connecting a motor, identifying a sensor, making a robot move forward, or testing a simple loop. These early wins matter because they reduce fear and create momentum.

A trainer should also normalize failure from the beginning. Robots rarely work perfectly on the first attempt. Wheels may be loose, wires may disconnect, code may be incomplete, and sensors may misread the environment. If failure is treated as embarrassment, learners withdraw. If failure is treated as data, learners become problem-solvers.

The language used by the trainer matters. Instead of asking, 'Who made a mistake?' the trainer can ask, 'What did the robot teach us?' This small change helps learners understand debugging as investigation rather than punishment.

3. What Learners Actually Gain from Robotics

The visible output of a robotics session is a moving robot, but the hidden learning is often more important. Learners develop computational thinking as they break a task into steps, sequence instructions, test conditions, and improve logic. They learn engineering thinking as they connect structure, stability, weight, friction, and movement. They learn communication as they explain why their robot behaves the way it does.

Teamwork is one of the strongest outcomes. Robotics naturally forces collaboration because one learner cannot do everything at once. Teams need builders, programmers, testers, observers, timekeepers, and presenters. When roles are well assigned, even shy learners find a way to contribute.

Robotics also develops resilience. A robot that fails to complete a task gives immediate feedback. Learners must troubleshoot patiently. They discover that improvement comes through repeated testing rather than quick perfection. This mindset is useful beyond STEM; it supports life skills, leadership, and academic confidence.

Another important gain is creativity. Learners often surprise trainers with unexpected designs, shortcuts, and strategies. When given enough freedom, they move beyond instructions and begin to innovate. This is why robotics should not be limited to copying manual steps. Manuals can introduce structure, but open-ended challenges create ownership.

Field insight: *The most important product of a robotics session is not the robot. It is the learner who leaves believing they can solve problems through design, testing, and teamwork.*

4. Common Field Challenges

The first major challenge is access. Robotics kits are expensive, and many schools cannot afford enough equipment for every learner. This means trainers must design group-based activities that keep all learners involved. Without careful role assignment, only one or two learners may handle the robot while others become spectators.

The second challenge is time. Robotics requires building, programming, testing, debugging, and reflection. A short session may excite learners but fail to produce deep understanding. Schools and partners should avoid treating robotics as a one-day performance activity. Sustainable learning requires repeated sessions.

The third challenge is teacher confidence. Some teachers support robotics enthusiastically but feel unsure about the technical side. Others may see it as separate from the curriculum. Trainers must therefore involve teachers, explain the learning value, and show how robotics connects to science, mathematics, ICT, design thinking, and communication.

The fourth challenge is classroom control. Robotics sessions are active and noisy. Learners move, discuss, test, argue, and celebrate. This energy is good, but it must be structured. Clear rules about kit handling, safety, turn-taking, and group roles are essential.

The fifth challenge is unequal participation. Boys may sometimes dominate technical tasks, while girls are pushed into note-taking or presentation roles. Trainers must actively prevent this by rotating roles and ensuring that every learner builds, programs, tests, and speaks.

5. The Role of the Trainer

A robotics trainer is not merely a technical instructor. The trainer is a facilitator, mentor, classroom manager, motivator, and translator of complex ideas into learner-friendly experiences. The best trainers do not simply give answers; they ask questions that help learners discover answers.

Good facilitation begins with clarity. Learners should know the challenge, the time available, the expected output, and the criteria for success. Ambiguity can create creativity, but too much ambiguity creates confusion. A trainer should provide enough structure to guide learners while leaving enough space for experimentation.

Observation is another key skill. A trainer must notice which team is stuck, which learner is being ignored, which group is rushing without testing, and which students need encouragement. Robotics facilitation requires movement around the room, listening to group conversations, and offering timely support.

The trainer must also protect the learning process. It is tempting to fix the robot for learners when time is running out. Sometimes this is necessary, but too much trainer intervention weakens ownership. A better approach is to guide with questions: What changed after the last test? Which part is not responding? What should we test first?

The trainer's attitude sets the culture. If the trainer is patient, curious, and calm, learners feel safe to experiment. If the trainer is impatient or overly focused on winning, learners may fear mistakes.

6. Girls in STEM and Inclusive Participation

Robotics has great potential to support girls in STEM, but inclusion does not happen automatically. In mixed groups, girls may be assigned supportive roles while boys handle building and programming. This pattern must be interrupted deliberately. Inclusion means ensuring that every learner touches the robot, writes or edits code, tests performance, and presents ideas.

One useful strategy is role rotation. During a session, each learner should experience different responsibilities: builder, programmer, tester, documenter, and presenter. This prevents one learner from controlling the entire process. It also helps learners discover strengths they did not know they had.

Representation matters as well. Trainers should use examples that show girls and women as engineers, programmers, inventors, and leaders. Learners need to see STEM as open to everyone. Encouragement should be specific: 'Your debugging approach was strong,' or 'Your design explanation was clear,' rather than generic praise.

Inclusive robotics also considers learners who are shy, slower to start, or less confident in English or technical vocabulary. A good trainer creates multiple ways to participate: speaking, building, drawing, testing, observing, explaining in simpler language, or recording results.

Ultimately, inclusive robotics is not about inviting learners into a room. It is about ensuring that they meaningfully participate once they are inside.

7. From Workshops to Competitions

Competitions can motivate learners and give robotics programs visibility. They encourage teamwork, strategy, discipline, and performance under pressure. They also expose students to peers from other schools and show them what is possible. However, competitions should not become the only purpose of robotics education.

When programs focus only on winning, weaker learners may be excluded and only the strongest team members may receive attention. This can turn robotics into an elite activity. A healthy program treats competitions as one part of a broader learning journey. Every learner should gain foundational skills before a smaller competition team is selected.

Preparation for competition requires more than technical practice. Learners need communication skills, documentation habits, time management, ethical behaviour, and emotional resilience. They must learn how to handle failure, respect other teams, explain design choices, and improve after each round.

Trainers should also help learners connect competition tasks to real-world problems. A robot challenge should not feel like an isolated game. It should lead to conversations about automation, engineering, local innovation, community needs, and future careers.

The best competition teams are not always the ones with the most advanced robot. They are often the teams that understand their design, communicate clearly, collaborate under pressure, and keep improving.

8. Recommendations for Sustainable Robotics Programs

First, schools and partners should start with clear learning goals. Robotics should be connected to curriculum, digital literacy, creativity, problem-solving, and career exposure. Without clear goals, programs may become impressive but shallow.

Second, trainers should design progressive learning pathways. Beginners need simple movement, sensors, loops, and teamwork activities. Intermediate learners can explore line following, obstacle avoidance, design improvement, and project documentation. Advanced learners can attempt competitions, innovation projects, and mentorship roles.

Third, schools should invest in teacher involvement. A robotics program is more sustainable when school teachers understand the tools and can support learners between external training sessions. Trainers should leave behind lesson outlines, activity guides, and maintenance practices.

Fourth, equipment should be managed carefully. Kits should be labelled, checked before and after sessions, and stored properly. Missing parts can weaken future sessions. Learners should be taught responsibility for shared resources.

Fifth, programs should document impact. Schools and partners should track participation, learner reflections, projects completed, gender balance, teacher involvement, and competition outcomes. Documentation helps improve the program and attract support.

Finally, robotics should be localized. Learners should be encouraged to design solutions around problems they recognize: transport, agriculture, waste, water, disability support, school safety, and community service. This makes robotics meaningful rather than abstract.

9. Conclusion

Robotics education is powerful because it brings together the mind, the hand, and the imagination. It allows learners to think, build, test, explain, and improve. In Kenyan schools, robotics can help young people see themselves as creators of technology and solvers of real problems.

The field experience is clear: the robot is important, but the learning culture is more important. A successful robotics session requires confidence, inclusion, structure, patience, and reflection. Learners need opportunities to fail safely, work in teams, ask questions, and connect technology to life beyond the classroom.

For schools, NGOs, trainers, and partners, the challenge is to move robotics from occasional excitement to sustained learning. When programs are well facilitated and inclusive, robotics becomes more than a STEM activity. It becomes a pathway to confidence, creativity, innovation, and future readiness.

Practical Session Checklist

Before session	Confirm kit parts, batteries, laptops, lesson objective, group roles, and safety rules.
During session	Rotate roles, observe participation, ask guiding questions, and encourage testing before conclusions.
After session	Run reflection, document learner feedback, inspect kits, and note improvements for the next lesson.

References

- Bers, M. U. (2020). Coding as a Playground: Programming and Computational Thinking in the Early Childhood Classroom. Routledge.
- Papert, S. (1980). Mindstorms: Children, Computers, and Powerful Ideas. Basic Books.
- Eguchi, A. (2014). Robotics as a Learning Tool for Educational Transformation. Proceedings of the International Conference on Robotics in Education.
- UNESCO. (2023). Technology in Education: A Tool on Whose Terms? Global Education Monitoring Report.
- OECD. (2023). Digital Education Outlook and related work on digital learning and skills.
- World Economic Forum. (2023). Future of Jobs Report: Skills, technology adoption, and workforce transformation.